Xiaoyu (Tom) Chen

Mandarin (Native) · English (Fluent) · Spanish · French (Intermediate) · German · Japanese · Portuguese (Beginner)
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Education

University of Michigan — Ann Arbor, MI
B.S. Computer Science (incoming transfer) — *Expected May 2028*

University of Florida — Gainesville, FL
B.S. Computer Science; GPA: 3.92/4.0 — *Aug 2024 – May 2026*

Research Experience

Citi × UF Industry Collaboration: TEA Lab, UF ECE — Gainesville, FL
Undergraduate Research Assistant, Dr. Ivan Ruchkin — *May 2026 – Present*

- Own the full data pipeline for a Citi-sponsored LLM agentic safety project (FinVault benchmark): built single-pass logprob/entropy extraction (Qwen3-8B), executed a 1,064-case run across 31 scenarios on dual-H200 HPC (SLURM); fixed silent CUDA corruption, an fp16 overflow bug, and a 100% case-ID label-leakage vulnerability via a verified anonymization pipeline
- Validated results at scale (Kruskal-Wallis $p<0.000001$, $n=1{,}064$); found defended cases show the highest model uncertainty, a resistance signature rather than confusion, presented directly to Citi stakeholders
- Investigated a suspiciously strong TRACES baseline (AUROC 0.987); built a diagnostic classifier proving it was a scenario-identity shortcut, not real detection, with accuracy collapsing to chance under scenario-grouped splitting

AdaptGuard, OrderLab (Systems Reliability) — University of Michigan EECS
Undergraduate Researcher, Prof. Ryan Huang; Supervised by Yuxuan Jiang — *Jul 2026 – Present*

- Identified a core failure mode in static agent-sandboxing while running controlled experiments across policy strictness levels on matched cheat/legitimate coding-agent scenario pairs: a permissive policy let an agent fake a bug fix by weakening a test assertion, while a strict policy blocking the cheat also blocked a legitimate regression-test addition; finding recognized by the PhD lead as a key insight in a project targeting OSDI publication, December 2026
- Sourced and quality-controlled two non-adversarial evaluation datasets (12 cases from Agent-SafetyBench, 14 from SABER) against PI-set criteria, spanning failure categories including privilege escalation and information leakage

Computer Vision and Sensing Lab, UF ABE — Gainesville, FL
Undergraduate Research Assistant, Dr. Henry Medeiros (volunteer) — *Feb 2026 – Aug 2026*

- Built a CV perception pipeline (depth estimation, YOLO, SAM3 segmentation) for a 20-person bamboo-harvesting robot team; trained/ran inference on HiPerGator

LLM Hate Speech Detection, University of Florida — Gainesville, FL
Research Participant — *May 2025 – Jul 2025*

- Designed an Elo-based scoring model converting binary human preference labels into a continuous 0–10 scale; integrated into LLaMA-4 and DeepSeek fine-tuning pipelines and annotated 450 dialogue responses for CRetor, a Mandarin counter-speech dataset

LLM Medical Dialogue, Zhongbo Zhichuang Technologies — Guangzhou, China
Independent Researcher — *Sep 2025 – Present*

- Combined VQ-VAE discrete latent encoding with doctor-patient dialogue corpora to improve semantic controllability and interpretability of medical dialogue generation; presented orally at AICCONF 2026 (IEEE SMC)

Publications

Xiaoyu Chen. "A Doctor Dialogue Model Based on Vector Quantisation." *AICCONF 2026* (IEEE SMC), Prague. Oral presentation.
Xiaoyu Chen. "From Theory to Reality: The Quest for Gravitational Waves." *CONF-CIAP 2024* (Theoretical and Natural Science; Scopus, CPCI).
Xiaoyu Chen et al. "CRetor: A Mandarin Counter-Speech Dataset Annotated for Rhetorical Strategy." Submitted to ACL Rolling Review (ARR), 2025.

Internship Experience

FlagIt — Gainesville, FL
Web Developer Intern — *Aug 2024 – Jan 2025*

- Built frontend UI with HTML/CSS and backend SQL schemas for user account data; collaborated with a team of 5 via Git

Chao Long Solar Technologies — Tianchang, Anhui, China
Engineering Intern — *Dec 2023 – Jan 2024*

- Analyzed photovoltaic manufacturing processes and proposed workflow improvements under engineer mentorship

Shanghai Howbuy Fund Sales Co., Ltd. — Shanghai, China
Design Department Intern — *Jul 2023 – Aug 2023*

- Produced 30 short-form videos reaching 100K+ viewers covering securities analysis and market indicator evaluation

Projects

Muted (YaBaYingYu) | *Next.js 14, Supabase, DeepSeek API, Web Speech API, Stripe* — 2026 – Present

- Solo-built an AI-first English speaking practice app for the Asian market end-to-end: persona-based conversation practice, real-time grammar correction, spaced-repetition vocabulary, and exam-specific practice modes (IELTS/TOEFL/Gaokao)
- Re-architected the voice pipeline after identifying OpenAI's Realtime API as cost-prohibitive at scale (∼$0.30/min); redesigned around browser-native Web Speech API plus DeepSeek text completion, cutting per-minute cost over 95% (∼$0.30 to $0.01–0.02)
- Found and fixed a privilege-escalation vulnerability allowing users to self-upgrade subscription tier via client-side database calls; implemented column-level permission controls to close the gap

Hoopalytics | *React, Vite, Django, Python, Tailwind CSS* — Sep 2025 – Nov 2025

- Self-implemented merge/heap sort comparison engine with a custom weighted NBA metric formula; 3-person team via GitHub

Technical Skills

Languages: Python, C++, JavaScript, TypeScript, SQL, HTML/CSS, Java
AI/ML: PyTorch, HF Transformers, OpenAI API, YOLO, SAM3, logprob extraction, 4-bit quant (NF4), STL anomaly detection
Infrastructure & Tools: Git, Docker, CI/CD, HiPerGator HPC, SLURM, RunPod, OpenRouter, DeepSeek API, Vercel
Frameworks: React, Next.js, Node.js, PostgreSQL, pandas, NumPy